

CYBER SECURITY INTERNSHIP – TASK 1

Understanding Cyber Security Basics & Attack Surface

1. Introduction to Cyber Security

Cyber security is the discipline of protecting computers, servers, mobile devices, networks, and data from malicious attacks. With the rapid growth of internet-based services such as online banking, social media, and cloud platforms, cyber threats have increased significantly. Cyber security ensures trust, privacy, and safe digital communication.

2. Importance of Cyber Security

Cyber security is essential to protect personal data, financial information, and organizational assets. Without proper security controls, attackers can steal sensitive information, disrupt services, and cause financial and reputational damage.

3. CIA Triad

3.1 Confidentiality

Confidentiality ensures that sensitive information is accessible only to authorized individuals. Techniques such as encryption, authentication, and access control are used to maintain confidentiality.

3.2 Integrity

Integrity ensures that data remains accurate, consistent, and unaltered during storage or transmission. Hashing, checksums, and digital signatures help maintain data integrity.

3.3 Availability

Availability ensures that systems and services remain accessible to legitimate users when required. Backups, redundancy, and protection against Denial-of-Service attacks support availability.

4. Types of Cyber Attackers

- Script Kiddies – Inexperienced attackers using publicly available tools.
- Insiders – Employees or trusted users abusing authorized access.
- Hacktivists – Attackers driven by political or social causes.
- Cyber Criminals – Attackers focused on financial gain.
- Nation-State Attackers – Highly skilled government-sponsored hackers.

5. Attack Surface

An attack surface refers to all possible points where an attacker can attempt to gain unauthorized access to a system. Reducing the attack surface is a key goal of cyber security.

Examples of attack surfaces include web applications, mobile applications, APIs, network ports, cloud storage services, and third-party integrations.

6. OWASP Top 10

The OWASP Top 10 is a globally recognized awareness document that lists the most critical security risks to web applications. It helps developers and security professionals understand common vulnerabilities.

- SQL Injection – Manipulating database queries through malicious input.

- Cross-Site Scripting (XSS) – Injecting malicious scripts into web pages.
- Broken Authentication – Weak login and session management.
- Security Misconfiguration – Incorrect system or server settings.
- Sensitive Data Exposure – Lack of proper encryption and protection.

7. Data Flow and Attack Points

In a typical application, data flows from the user to the application, then to the server, and finally to the database. Each stage is a potential target for attackers.

User-level attacks include phishing and social engineering. Application-level attacks include XSS and CSRF. Server-level attacks involve misconfigurations, while database-level attacks include SQL injection.

8. Conclusion

A strong understanding of cyber security fundamentals such as the CIA triad, attacker types, attack surfaces, and OWASP Top 10 vulnerabilities is essential for building secure systems and defending against cyber threats. This knowledge forms the foundation for advanced cyber security learning and practical implementation.